Geist in the Machine — Volume Spine

A Research Atlas of Dialectical Agent Pedagogy

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Orientation

A research programme testing whether an AI tutor teaches better when you change its *inner structure* — not the model it runs on — and sorting which of those structures genuinely help from the ones that only look like they do.

Most AI tutors are a single model answering a single prompt. This work gives the tutor an *interior* and watches what changes: a part that drafts a reply, a second part that criticises that draft before the learner ever sees it, and an orientation that treats the learner as a thinking subject to be met rather than a deficit to be filled. The wager is that this internal architecture — dialogue, self-critique, recognition — produces better teaching than raw capability alone.

The atlas is organised around a plain result: some of the architecture works, some of it is redundant, and some of it only looks like it works. Read this page for the whole story in plain terms; open any module below for the detail; or take the consolidated paper for the entire argument, end to end.

*This sentence is the only one actually authored by Liam Magee in this paper.

The core idea

An interior for the tutor. Two ideas, both borrowed from philosophy, do the heavy lifting — and both are easier to grasp from the everyday version first.

The first is *self-critique*. Instead of a tutor that says the first thing it generates, picture one with two voices: a drafter that proposes a reply, and an internal critic that reads the draft and asks "is that actually sound? what is it missing?" before anything reaches the learner. We borrow Freud's names — the *ego* (the drafter, which keeps final say) and the *superego* (the critic) — but the mechanism is simply structured self-revision. The same structure can be given to the simulated learner, so both sides of the dialogue deliberate before they speak.

The second is *recognition*. A tutor can treat a learner as a container to be filled, or as a thinking subject whose reasoning — including the mistakes — is worth meeting on its own terms. The second stance comes from Hegel's account of *mutual recognition*: one becomes a full subject by being recognised as one. Turned into a prompt, recognition means eliciting and building on the learner's own thinking rather than overwriting it. A central question is whether the philosophical *vocabulary* matters, or whether it is the plain orientation underneath that works.

Holding both ideas lets us ask precise questions. Does the critic catch real errors, or just add words? Does recognition change the teaching, or only the tone? Does modelling the learner's hidden state help, or is the appearance of "understanding the learner" an artifact of how the dialogue was logged? The three families below are the three answers.

The findings

The findings sort into three families. Each carries a *claim-status* — a one-word commitment to how firm the result is (the full grammar is further down). Here is the shape of the results before the detail.

Recognition works like calibration

[**SETTLED**] Settled within current evidence

Recognition-oriented prompting behaves like *calibration*: it lifts a tutor's weakest dimensions the most and narrows the variation within a response — raising the floor more than the ceiling. It works with or without the separate critic, and — the part that surprised us — it works just as well when the Hegelian vocabulary is stripped out and replaced with a plainly-worded pedagogical orientation. The active ingredient is the stance, not the theory-words. The critic, for its part, does catch and correct real errors.

Caveat. The critic's *extra* benefit beyond recognition is model-dependent: large on weaker models, close to zero on strong ones. On a capable model, recognition alone captures most of the gain.

Drama and charisma — voice as an instrument

[**EXPLORATORY**] Exploratory

A parallel line gives the tutor a charismatic, performed voice — a durable "id" that repeatedly authors a fresh speaking persona — and scores it on a separate charisma rubric. It can produce an extraordinary pedagogical presence. The open question is whether *productive internal conflict* can be engineered as a stable source of teaching quality, rather than merely role-played.

Caveat. Charismatic voice trades off against recognition-style elicitation — the most charismatic tutors are not the best at drawing the learner out. This family is an active arc, not a settled result; its main section is scaffolded, with prose still landing in the paper.

A rich learner-interior did not help

[**KILLED**] Killed / closed under a specified gate

The deepest line asks whether the tutor benefits from carrying a rich, explicit model of the learner's hidden interior — externalised state, theory-of-mind elaborations, counterfactual replay of what a different move would have done. The answer, repeatedly, was no: these elaborations came out null against a lean baseline that simply lets the model infer the learner implicitly.

Caveat. An early apparent win for the structured architecture turned out to be a turn-counting artifact; once it was fixed and the baseline re-run, the gap fell to within noise. The durable contribution from this family is the *method* — the trap scenarios and the adaptive runner — not a modelable interior.

How we know it

A claim about a mechanism is only as good as the apparatus that tests it, and several pieces of that apparatus are contributions in their own right.

Provable discourse. Every mechanism claim in the paper is machine-checked against the evaluation database before it is allowed to stand: the number in the prose and the number in the data must agree, or the build complains. This atlas is generated from that same source, so it cannot quietly drift from the evidence.

Process tracing. Rather than only scoring the final answer, the apparatus records the tutor's internal steps — draft, critique, revision — so we can ask not just whether a method helped but *why*, and watch a critique get incorporated turn by turn.

Rubrics as instruments. The scoring rubric went through four iterations, and we treat that evolution as construct development — working out what "good tutoring" even means to measure — rather than incidental tooling. The same rubric doubles as a signal for automatically tuning prompts, with gains that transfer unevenly across models.

Evidence map

The modules below are slices of the one paper, each tagged with its claim-status. Filter them by status, or open any one as a self-contained PDF. The grid is generated from the manifest, so a status change propagates here automatically.

#	Module	Family	Status
1	Recognition as Calibration	Recognition and Pedagogical Calibration	SETTLED
2	Superego as Conditional Error Corrector	Recognition and Pedagogical Calibration	SETTLED
3	The Adaptive-Responsiveness Null	Recognition and Pedagogical Calibration	KILLED
4	Provable Discourse and Process Tracing for Agent Architectures	Apparatus as Method	METHODS
5	Drama Machine Pedagogy	Drama and Charisma	PLANNED
6	The Id-Director and Charismatic Pedagogy	Drama and Charisma	EXPLORATORY
7	Ontology, State, and the Failure of Rich Learner Interiors	Ontology, State, and the Limits of Interior Modeling	KILLED
8	Rubrics as Instruments and Optimization Signals	Apparatus as Method	METHODS
9	Human Learning Translation Agenda	Translation and Future Programme	FUTURE
10	Model Dependence and Mechanism Boundary Conditions	Recognition and Pedagogical Calibration	SETTLED
11	Teaching-Drama Axiom Transfer	Drama and Charisma	SCOPE-BOUND

The modules

Module 1 — Recognition as Calibration

[**SETTLED**] Settled within current evidence · *Recognition and Pedagogical Calibration*

The clean prompt-level mechanism: recognition acts as calibration, a floor-lifting and variance-narrowing transformation of tutor output. Includes the A10/A10b matched-pedagogy controls that locate the effect at the orientation-family level.

Sources: §3.2, §6.1, §7.9, §7.10

Module 2 — Superego as Conditional Error Corrector

[**SETTLED**] Settled within current evidence · *Recognition and Pedagogical Calibration*

The architecture-level mechanism: what the superego catches, how critique resolves across deliberation rounds, the factorial interaction, and the ~15–17% additivity deficit consistent across three models.

Sources: §6.2, §6.4, §7.3

Module 3 — The Adaptive-Responsiveness Null

[**KILLED**] Killed / closed under a specified gate · *Recognition and Pedagogical Calibration*

The third pre-registered mechanism, turned from a failed hypothesis into a substantive well-powered null — including the insight-action gap and the A16 cumulative-rewrite pre-registration.

Sources: §6.3

Module 4 — Provable Discourse and Process Tracing for Agent Architectures

[**METHODS**] Methods / apparatus contribution · *Apparatus as Method*

The apparatus-as-method module: superego-critique taxonomy, the provable-discourse framework that verifies every mechanism claim against data, and the reproducibility chain.

Sources: §5.3, §5.9, §5.11, §7.4

Module 5 — Drama Machine Pedagogy

[**PLANNED**] Planned — scaffold, prose pending · *Drama and Charisma*

An active arc. Develops deliberative tension, role differentiation, and anti-sycophantic friction as a pedagogical architecture extending — but not reducible to — the ego/superego result. Prose lands as a new section of the canonical paper; this module is the scaffold.

Sources: — (scaffold; prose pending)

Module 6 — The Id-Director and Charismatic Pedagogy

[**EXPLORATORY**] Exploratory · *Drama and Charisma*

A parallel architecture family — a transient ego repeatedly authored by a durable id — scored on a Weberian charisma rubric independent of the v2.2 tutor rubric.

Sources: §6.7, §8.9

Module 7 — Ontology, State, and the Failure of Rich Learner Interiors

[**KILLED**] Killed / closed under a specified gate · *Ontology, State, and the Limits of Interior Modeling*

The deepest theoretical arc, mostly negative: the adaptive runner and trap methodology, bilateral-ToM and state-schema ablations, hypothesis tracking, and the concealed-interior signal test. Asks whether there is a modelable interior at all, or whether interiority is an artifact of trace design.

Sources: §6.8, §6.9, §6.10

Module 8 — Rubrics as Instruments and Optimization Signals

[**METHODS**] Methods / apparatus contribution · *Apparatus as Method*

Treats the four rubric iterations, the measurement paradox, scoring, and automated prompt tuning as instrument design rather than incidental tooling.

Sources: §5.2, §5.6, §7.7, §7.8

Module 9 — Human Learning Translation Agenda

[**FUTURE**] Future empirical agenda · *Translation and Future Programme*

Separates judged tutor quality from human learning outcomes, motivated by the model-dependent tutor-learner asymmetry and the synthetic-learner / LLM-judge limits. The translation programme and its pilot infrastructure.

Sources: §6.5, §8.1, §8.2

Module 10 — Model Dependence and Mechanism Boundary Conditions

[**SETTLED**] Settled within current evidence · *Recognition and Pedagogical Calibration*

The cross-cutting boundary-condition result that qualifies both calibration and error correction. Base-tutor capability gaps across the three-model triad, the model-dependent superego residual, the Writing-Pad ablation showing recognition operates at the prompt level rather than through a Freudian memory pad, and the six-model capability-threshold test that does not find the helpful-then-harmful pattern the prosthesis hypothesis predicted.

Sources: §5.7, §5.8, §6.6, §8.3

Module 11 — Teaching-Drama Axiom Transfer

[**SCOPE-BOUND**] Supported but scope-bound · *Drama and Charisma*

A scope-bound A19 module. The scaffold defines a protocol, literature-positioning matrix, card-level verdict grammar, attempt-1 materializer, blind-adjudication scaffold, and sidecar-paper outline. Section 7.9 is the only current canonical source: A18's 10-of-14 bounded result remains the primary evidence, while A19 now adds a local one-family screen with two n=1 surface-agreement headroom candidates followed by a two-seed stability smoke that fails to replicate either candidate. No pooled A19 rate, stability-confirmed A19 transfer effect, human-learning, deployment, or weight-learning claim is licensed.

Sources: §7.9

Claim-status grammar

- [**SETTLED**] — Settled within current evidence
- [**SCOPE-BOUND**] — Supported but scope-bound
- [**EXPLORATORY**] — Exploratory
- [**KILLED**] — Killed / closed under a specified gate
- [**SPECULATIVE**] — Speculative / theoretical
- [**METHODS**] — Methods / apparatus contribution
- [**PLANNED**] — Planned — scaffold, prose pending
- [**FUTURE**] — Future empirical agenda

A short glossary

Ego / superego. Two internal voices given to the tutor (and optionally the learner): an *ego* that drafts and keeps final say, and a *superego* that critiques the draft first. Borrowed from Freud as names; mechanically, structured self-revision.

Recognition. Treating the learner as a thinking subject whose reasoning is worth meeting, not a gap to fill. From Hegel's mutual recognition; in practice, a prompt orientation that elicits and builds on the learner's own thinking.

Calibration. How recognition acts on tutor output: lifting the weakest dimensions and narrowing the variation — raising the floor more than the ceiling.

Claim-status. A one-word commitment to how firm a result is — settled, scope-bound, exploratory, killed, and so on. The full grammar is in the status section.

Trap suite. A set of scenarios engineered to tempt a tutor into a specific pedagogical mistake, used to test whether an architecture changes its move when the situation demands it.

Provable discourse. The discipline of machine-checking every empirical claim in the paper against the evaluation database, so the prose and the data cannot disagree.

Id-director. The drama/charisma architecture: a durable "id" that repeatedly authors a fresh, performed tutor persona, scored on a separate charisma rubric.

What's still open

Two limits bound everything above, and they matter as much as the results.

First, every finding here concerns *tutor output quality as judged by a language model*, interacting with a *simulated* learner. Whether the supported mechanisms — recognition-as-calibration, conditional error correction — improve actual *human* learning is open, and is the next programme rather than a result of this one. A human-learner pilot is built and waiting on the usual approvals.

Second, the magnitudes are bounded by model capability. The mechanisms hold across capable models, but how much they buy you scales with the underlying model — the critic's residual value, in particular, can shrink toward nothing on a strong enough model. None of this is a deployment claim or a weight-learning claim; it is a set of carefully scoped findings about agent architecture, with the boundaries marked.